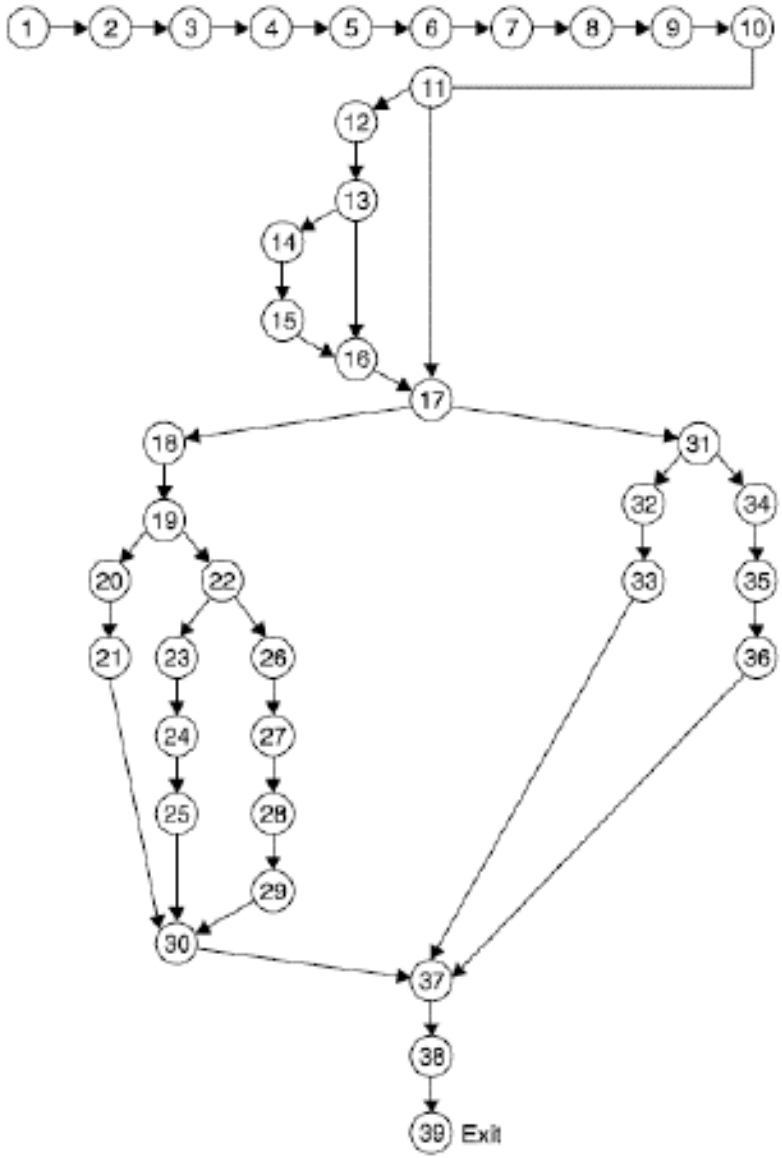
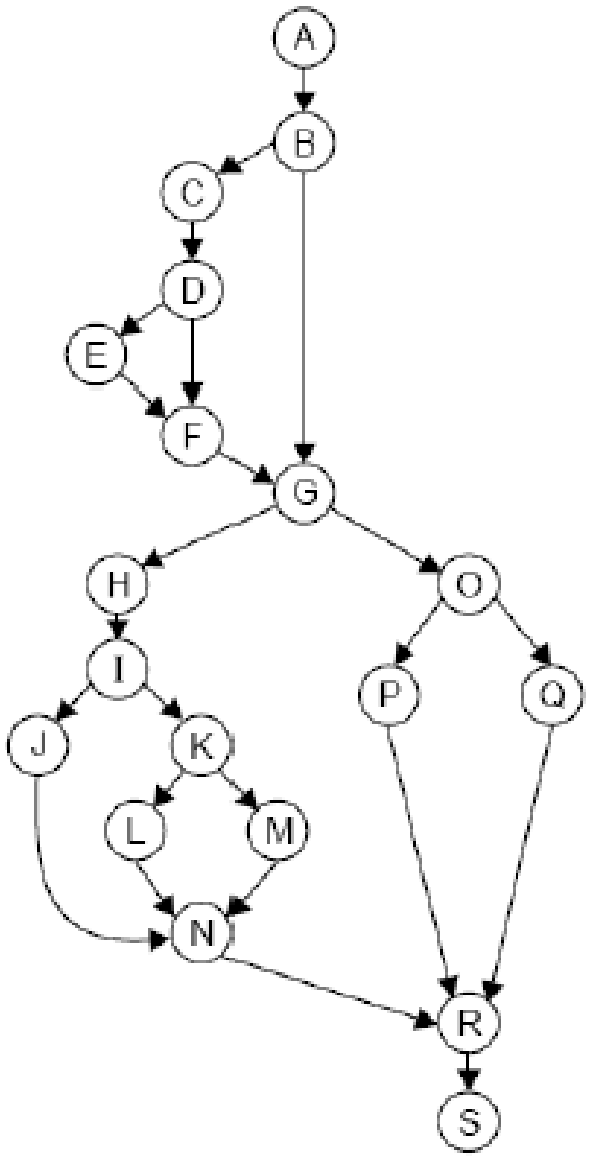
Software Testing



Program flow graph



DD Path graph (Decision to decision path graph)

Multiple condition/decision coverage (MC/DC)

- It ensures Decision as well as condition coverage.
- *A test suite would achieve MC/DC if during execution of the test suite each condition in a decision expression independently affects the outcome of the decision.*
- Requirements:
  1. Every decision expression in a program must take both true as well as false values.
  2. Every condition in a decision must assume both true and false values.
  3. Each condition in a decision should independently affect the decision's outcome.

  Number of test cases required would be generally (no. of conditions +1)

if((A&&B)||C)

Test Case	ABC	ReFsult	A	B	C
1	TTT	T	5		
2	TTF	T	6	4	
3	TFT	T	7		4
4	TFF	F		2	3
5	FTT	F	1		
6	FTF	F	2		
7	FFT	F	3		
8	FFF	F			

A	[1,5]	[2,6]	[3,7]
B	[2,4]		
C	[3,4]		

[2,3,4,6] or
[2,3,4,7] or
[1,2,3,4,5]

```
if((A||B)&&C)
```

If ((A&B)&(C||D))

Draw the control flow graph, compute cyclomatic complexity and basis paths

```
#include <stdio.h>
int main()
{
int a, b, c, max;
printf("\nEnter 3 numbers");
scanf("%d %d %d", &a, &b, &c);
max=a;
if(b>max)
  max=b;
if(c>max)
  max=c;
printf("Largest No is %d", max);
return 0;
}
```

Draw the control flow graph, compute cyclomatic complexity and basis paths

```
#include <stdio.h>
int main()
{
int n,r;
printf("\nEnter the number");
scanf("%d", &n);
r=n%2;
if(r==0)
  printf("EVEN");
else
  printf("ODD");
return 0;
}
```